

Superstore Sales & Customer Analytics Dashboard

End-to-End SQL + Power BI Analytics Project

1. Project Overview

This project focuses on analyzing Superstore sales and customer data using MySQL and Power BI. The workflow included importing raw CSV data into MySQL, performing data cleaning and validation, executing SQL-based analysis, and building interactive dashboards in Power BI. The project provides insights into:

- Sales performance
- Customer behavior
- Category-wise revenue
- Regional trends
- Profitability analysis

2. Tools Used

Tool	Purpose
MySQL	Data Cleaning, Validation & Analysis
Power BI	Dashboard Creation & Visualization
Excel / CSV	Dataset Source

3. Data Cleaning & Validation

The following data preparation steps were performed:

- Imported CSV dataset into MySQL
- Corrected date format issues
- Verified row counts
- Checked for NULL values
- Identified duplicate records
- Validated numeric and date data types

4. Important SQL Queries

Total Sales

```
SELECT SUM(sales) AS total_sales FROM orders;
```

Category-wise Revenue

```
SELECT category, SUM(sales) AS revenue
FROM orders
GROUP BY category
ORDER BY revenue DESC;
```

Monthly Sales Trend

```
SELECT MONTH(order_date) AS month,
SUM(sales) AS total_sales
FROM orders
GROUP BY MONTH(order_date);
```

Top Customers

```
SELECT customer_name, SUM(sales) AS total_sales
FROM orders
GROUP BY customer_name
ORDER BY total_sales DESC
LIMIT 10;
```

5. Dashboard Features

The Power BI dashboard includes: • KPI Cards (Sales, Profit, Orders, Customers) • Monthly Sales Trend Analysis • Customer Segment Contribution • Top Customers Analysis • Category-wise Sales Distribution • Customer Performance Table • Interactive Filters and Slicers

6. Key Insights

- Technology category generated the highest revenue contribution.
- Consumer segment contributed maximum sales.
- A small group of customers generated significant revenue.
- Sales showed seasonal fluctuations across months.
- Some transactions generated negative profit.
- Regional sales performance varied significantly.

7. Conclusion

This project demonstrates a complete end-to-end data analytics workflow using MySQL and Power BI. The project involved data cleaning, validation, SQL analysis, KPI creation, and dashboard storytelling. The final dashboard can help businesses monitor sales trends, identify valuable customers, and support data-driven decision-making.